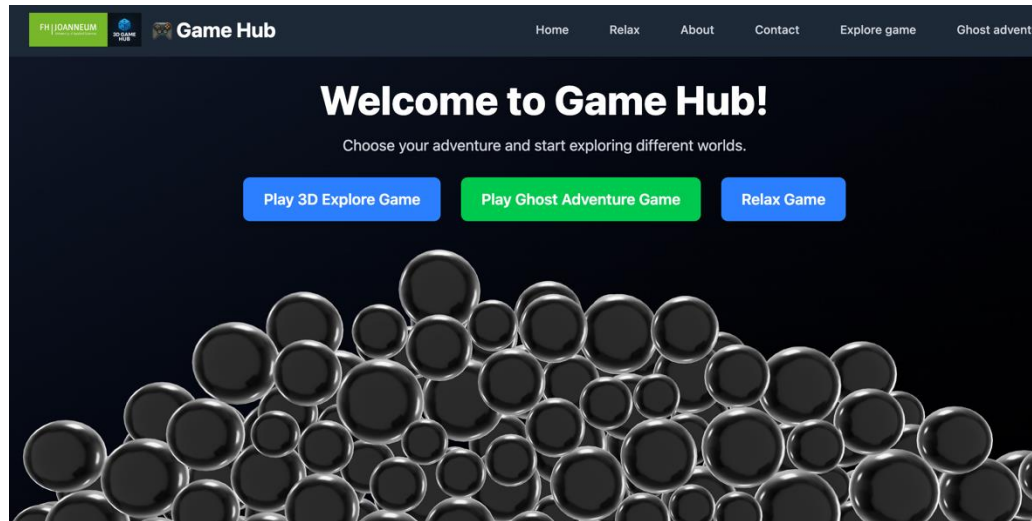




Game Hub 3D

- Mariia Glushenkova
- 3D Programming
- FH Joanneum

Project Overview



Interactive 3D Tool



A browser-based hub
showcasing interactive 3D
scenes



Tech Stack: React, Three.js,
TailwindCSS



Demonstrate modular scene
design, interactive navigation,
and 3D web rendering

Assets & References



Custom Assets: Basic geometries, shaders, UI components



Third-Party Assets: Public models and textures (see README)



References: Google Maps API, Three.js examples, official docs

Features

3D Map Explorer

Ghost Adventure Game

Meditation Scene

Player Selection UI

Rain Visual Effect



Built with:

- React + Vite
- @react-three/fiber, drei, rapier
- Google Maps JavaScript API
- Open-Meteo API
- Tailwind CSS

Highlighted Technique – Google Maps Integration & Stars Rendering

- Embed Google Maps in 3D scene and render animated stars

Google Maps:

```
const iframe = document.createElement('iframe'); iframe.src =  
`https://www.google.com/maps/embed?...`; iframe.width = "600";  
iframe.height = "450"; document.body.appendChild(iframe);
```

- Embeds a live Google Map inside the Three.js scene or UI panel

Stars:

```
for (let i = 0; i < 200; i++) { const star = new THREE.Mesh(geometry,  
material); star.position.set(...); scene.add(star); }Creates hundreds of  
particles to simulate a starfield background
```



Welcome to the Exploration Game!

Travel to real-world cities and collect stars!

Ghost

Robot

Enter a city or country

Start Game

Demo (Three.js)

- Link: <https://game-hub-3d.vercel.app/>
- Demo: [Screen Recording 2025-06-11 at 22.37.06.mov](#)

Final Notes



Repository Link:

<https://github.com/maguitaria/game-hub-3d>



Slides Location: Committed in repo